iPhone 15 Pro Repair Manual Summary

Compiled from Apple technical specifications and iFixit community repair data | Models: A2848, A3101, A3102, A3104 | August 2026

Note: Apple does not publish a full internal service manual for iPhone like traditional laptop manufacturers do. This summary is compiled from Apple's public tech specs and iFixit's published teardown and repair guides. For step-by-step photos and video, refer to ifixit.com.

1. SYSTEM SPECIFICATIONS

Chip & Performance

- Apple A17 Pro chip (3nm)
- 6-core CPU (2 performance + 4 efficiency cores)
- 6-core GPU, 16-core Neural Engine
- 8GB RAM

Display

- 6.1" Super Retina XDR OLED, 2556 x 1179 at 460 ppi
- ProMotion adaptive refresh up to 120Hz, Always-On display
- 1000 nits typical / 2000 nits peak outdoor brightness
- Dynamic Island, Ceramic Shield front cover

Battery & Storage

- Li-Po, 3274 mAh (approx. 17.1 Wh)
- Storage options: 128GB / 256GB / 512GB / 1TB
- USB-C (USB 3 support on Pro models), up to 20W wired fast charge, MagSafe/Qi wireless charging

Build

- Titanium frame, textured matte glass back
- IP68 water/dust resistance (6m, 30 min)
- Weight: 187g | 146.6 x 70.6 x 8.25 mm

2. SAFETY INSTRUCTIONS

Before Starting Any Repair

- Power off completely (hold side button + volume button, slide to power off)
- Disconnect from any charging cable
- Let battery drain below 25% before opening the device — a charged lithium-ion battery can catch fire if damaged
- Work on a clean, flat, ESD-safe surface with a grounding mat/wrist strap

Battery Handling Warning: Never puncture, bend, crush, or apply excess pressure to the battery. If a battery appears swollen, discontinue use immediately and handle with caution — do not attempt DIY removal of a swollen cell without proper precautions.

Screen/Glass Precautions

- If the screen or back glass is cracked, lay overlapping strips of packing tape over it before disassembly to contain shards
- Use a heating pad/heat gun (approx. 80–90°C) to soften adhesive before prying — excess heat can damage the OLED panel or battery
- Mark opening picks 3mm from the tip to avoid inserting too deep and damaging internal components

3. REQUIRED TOOLS

- P2 Pentalobe screwdriver (case screws)
- Phillips #000 screwdriver
- Tri-point Y000 screwdriver (logic board / internal brackets)
- Suction cup handle + opening picks / spudger
- Heating mat or heat gun (adhesive softening)
- Isopropyl alcohol (99%) for adhesive prep
- Tweezers, plastic prying tools, plastic cards

4. COMPONENT REMOVAL ORDER

For Battery Removal:

Pentalobe screws → Front display assembly (opened like a book from left edge) → Battery adhesive release tabs → Battery

For Screen Replacement:

Pentalobe screws → Heat/pry open display → Disconnect display + battery connectors → Remove display

For Charging Port / USB-C Assembly:

Pentalobe screws → Display → Battery → Speaker assembly → Charging port flex cable

For Logic Board Access:

Pentalobe screws → Display → Battery → Rear camera bracket → Various flex cable connectors → Logic board shield screws → Logic board

5. KEY COMPONENT REMOVAL STEPS

Case Opening

- Remove two 7mm Pentalobe screws at the bottom edge, next to the charging port
- Apply light heat along the display edges to soften adhesive
- Attach suction cup near the bottom edge and gently lift while sliding an opening pick around the perimeter
- Open like a book — front assembly is hinged, not fully detached

Battery

- Disconnect battery connector from logic board first, before any other work
- Battery is secured with stretch-release adhesive tabs (pull tabs) rather than only screws

- Apply isopropyl alcohol under the battery to help dissolve adhesive if tabs fail
- Do not pry or bend the battery cell during removal

Display Assembly

- Disconnect display data cable and battery connector before fully separating
- Screen and back glass are both replaceable; front/back opening design allows either-side access
- Original Apple screens require Repair Assistant/software calibration after replacement (True Tone, Face ID pairing)

Rear Camera System

- Secured with a small bracket and Phillips screws
- Note: camera modules on iPhone 15 series are part-paired to the logic board — swapping between units without Apple's configuration tool can cause errors or reduced functionality

USB-C Charging Port

- Accessed after removing display and battery
- Secured with small Phillips screws and a metal bracket
- Uses a plastic spudger to disconnect flex cable without tearing

6. PORTS & CONNECTIVITY

- USB-C connector (USB 3, up to 10 Gbps on Pro models)
- No physical SIM slot on US models — eSIM only
- MagSafe magnetic wireless charging array
- Qi2 wireless charging support
- 5G (sub-6GHz and mmWave on supported models), Wi-Fi 6E, Bluetooth 5.3

7. DIAGNOSTICS & TROUBLESHOOTING

Recovery Mode

- Connect to a computer or use a USB-C cable
- Press and quickly release Volume Up, then Volume Down, then press and hold the Side button until the recovery screen appears
- Restore or update via Finder (Mac) or iTunes (Windows)

DFU (Device Firmware Update) Mode

- Connect to computer, press and quickly release Volume Up, then Volume Down
- Press and hold Side button until screen goes black
- While holding Side button, also hold Volume Down for 5 seconds
- Release Side button, continue holding Volume Down for up to 10 seconds until device is recognized (screen stays black — this confirms DFU mode)

Force Restart

- Press and quickly release Volume Up
- Press and quickly release Volume Down
- Press and hold the Side button until the Apple logo appears

Common Troubleshooting

- Wi-Fi/network issues: toggle Airplane Mode, restart device, or reset network settings under Settings > General > Transfer or Reset
- Unresponsive touch/display: force restart first before assuming hardware fault
- Battery drain/swelling: check Settings > Battery > Battery Health; discontinue use if physically swollen

8. SCREW & FASTENER REFERENCE

Location	Screw Type	Notes
Bottom edge (case)	P2 Pentalobe, 7mm	2 screws next to charging port
Internal brackets	Tri-point Y000	Logic board shields, small brackets
Camera bracket	Phillips #000	Small bracket over rear camera flex
Charging port bracket	Phillips #000	Secures USB-C flex assembly
Battery	Stretch-release adhesive	No screws — pull-tab adhesive strips

9. IMPORTANT NOTES

Parts Pairing

iFixit's teardown testing found several iPhone 15 series components — including camera/LiDAR modules — are electronically paired to the original logic board. Swapping these parts between two units without going through Apple's official System Configuration process can cause the Camera app to crash or features to be disabled. This significantly limits independent, non-Apple-authorized repair options.

Repairability

iFixit gave the iPhone 15 series a provisional repairability score of 4 out of 10, citing the parts-pairing restrictions despite improved front-and-back opening access compared to older single-side-opening designs.

Repair Assistant / Calibration

- Genuine Apple screen and battery replacements should be calibrated using Repair Assistant (on-device) to restore True Tone and full functionality
- Face ID may show warnings if the display, battery, or Face ID module is replaced without proper Apple calibration

10. GETTING HELP

- Apple Support: support.apple.com or the Apple Support app
- Apple Store / Apple Authorized Service Provider for warranty repairs
- iFixit.com for community repair guides, tools, and replacement parts
- Self Service Repair program (Apple) for genuine parts and tools on eligible models

QUICK REFERENCE CHECKLIST

Before Starting Any Repair

- Back up device (iCloud or computer)
- Charge below 25% or note battery level
- Power off completely and disconnect cable
- Have correct screwdrivers ready (Pentalobe P2, Phillips #000, Tri-point Y000)
- Prepare heating mat/heat gun and isopropyl alcohol for adhesive

After Completing Repair

- Reconnect all flex cable connectors before closing
- Reinstall all screws in correct locations
- Run Repair Assistant / calibration if display or battery replaced
- Power on and test Face ID, cameras, Wi-Fi, and charging
- Check Settings > Battery > Battery Health after battery replacement

Sources: Apple Support (support.apple.com/en-us/111829), iFixit.com repair guides and teardown reports. This is a community/public-source compilation, not an official Apple service manual.